# TryHackMe_Intro_to_LAN.md

This room introduces the fundamental concepts powering Local Area Networks (LANs), including network topologies, routing vs. switching, subnetting principles, address resolution using ARP, and dynamic IP address allocation using DHCP.

Task 1: Introducing LAN Topologies

Local Area Networks (LANs) connect computing devices within a limited geographical area, such as an office or home.

- Switches vs. Routers: Switches centrally connect multiple local devices within the same network, while routers connect different networks together and handle packet routing across multiple paths.

- Network Topologies: Network architecture determines efficiency and fault tolerance. Bus or star topologies are generally cost-efficient to deploy, whereas mesh topologies provide high redundancy but are expensive to set up and maintain.

Questions & Answers:

1. What does LAN stand for? \rightarrow Local Area Network

2. What is the verb given to the job that Routers perform? \rightarrow Routing

3. What device is used to centrally connect multiple devices on the local network and transmit data to the correct location? \rightarrow Switch

4. What topology is cost-efficient to set up? \rightarrow Bus

5. What topology is expensive to set up and maintain? \rightarrow Mesh

6. Complete the interactive lab attached to this task. What is the flag given at the end? \rightarrow THM{...} (Retrieved after completing the interactive topology breaking exercise)

Task 2: A Primer on Subnetting

Subnetting involves segmenting a larger network into smaller, logically isolated sub-networks to improve efficiency, security, and administrative control.

Key Components:

- Network Address: Marks the beginning of the network range (e.g., 192.168.1.0).
- Host Address: Assigned to individual endpoints on the subnet (e.g., 192.168.1.100).
- Default Gateway: Forwards traffic destined for external networks (e.g., 192.168.1.254).
- Subnet Mask: A 32-bit value divided into 4 octets (ranging from 0 to 255) used to differentiate the network portion from the host portion.

Questions & Answers:

1. What is the technical term for dividing a network up into smaller pieces? \rightarrow Subnetting
2. How many bits are in a subnet mask? \rightarrow 32
3. What is the range of a section (octet) of a subnet mask? \rightarrow 0-255
4. What address is used to identify the start of a network? \rightarrow Network Address
5. What address is used to identify devices within a network? \rightarrow Host Address
6. What is the name used to identify the device responsible for sending data to another network? \rightarrow Default Gateways

Task 3: Address Resolution Protocol (ARP)

ARP maps logical network addresses (IP addresses) to physical hardware identifiers (MAC addresses) across a local link.

* Workflow:

- ARP Request: A broadcast message (FF:FF:FF:FF:FF:FF) asking "Who has IP address 192.168.1.10?"
- ARP Reply: A unicast response from the target device returning its physical MAC address.
- ARP Cache: A local lookup table storing resolved IP-to-MAC mappings.

Questions & Answers:

1. What does ARP stand for? \rightarrow Address Resolution Protocol

2. What category of ARP Packet asks a device whether or not it has a specific IP address? \rightarrow Request

3. What address is used as a physical identifier for a device on a network? \rightarrow MAC Address

4. What address is used as a logical identifier for a device on a network? \rightarrow IP Address

Task 4: Dynamic Host Configuration Protocol (DHCP)

DHCP automatically assigns dynamic IP configurations to client devices joining a network.

DORA Process:

- Discover: Client broadcasts to locate active DHCP servers.
- Offer: Server offers an available IP configuration (e.g., 192.168.1.10).
- Request: Client formally requests the offered IP lease.
- ACK (Acknowledge): Server acknowledges and confirms the lease duration.

Questions & Answers:

1. What type of DHCP packet is used by a device to retrieve an IP address? \rightarrow DHCP Discover

2. What type of DHCP packet does a device send once it has been offered an IP address by the DHCP server? \rightarrow DHCP Request

3. Finally, what is the last DHCP packet that is sent to a device from a DHCP server? \rightarrow DHCP ACK